Doc. 313 – No Beginning: The Thermal History on the Time Cycle

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Contents

Doc. 313 – No Beginning

What this is about

Doc. 312 ends with obligation D(ii): can the field configurations close periodically on the compact time cycle ($\tau_c = 2\pi/H_0 = 91.9$ Gyr) *while* carrying the observed thermal sequence – nucleosynthesis, recombination, structure formation? A circle has no beginning; the observations show a gradient. This document computes the question (`ffgft_313_zeitzyklus_geschichte.py`, `ffgft_313_fraktale_wegkorrektur.py`) and finds three results and one hard open core:

1. **Antipode finding [K|P20]:** the observable history fills *one half-cycle*. With the fractal path correction (Ch. B, independently from A040) the particle horizon sits within $0.14\,\%$ of the antipode ($\theta = 0.9986\,\pi$; smooth: $1.012\,\pi$), the scattering wall at $0.979\,\pi$ – the "beginning" is the antipode of the time cycle, not an event.
2. **Smooth closure [K]:** the observed mass-drift profile reaches the antipode with slope $\sqrt{\Omega_r} \approx 0.01$ – the smoothness condition of an even periodic closure is met to $1\,\%$, and the return half is exactly the photonically unobservable half.
3. **Entropy obstruction [K]:** exact τ_c periodicity is not dynamically generic (Poincaré discrepancy $\sim 10^{10^{104}}$); it must be imposed as a boundary condition. Fine-grained admissible, coarse-grained open – the sharpened core of D(ii).

Not part of the A-series; registration in Doc. 190 separately. References: Doc. 306 ($\partial \mathrm{T}^4 = \emptyset$), 267 (degeneracy), 312 (closure scale, compact time direction, statics of the relation, torus rest frame).

Chapter A

No boundary, no $t = 0$: the starting position

FFGFT's position on singularity and big bang stands since Doc. 306: $\partial \mathrm{T}^4 = \emptyset$ – the torus has no boundary, "what was at $t = 0$?" is wrongly posed. Doc. 312 made this concrete and gave it three supports:

Geometric: the time direction is a cycle of length 91.9 Gyr. A closed circle has no starting point – "$t = 0$" is as meaningless on it as "the beginning of the equator".

Backwards: in the mass-drift reading (Doc. 312, statics of the relation; Wetterich), running back does not send distances to zero but masses: $\tilde{T} \cdot m = 1$ then means $\tilde{T} \to \infty$ – no density divergence, but a regime of ever slower clocks. The "singularity" is where the clocks stand still, not where the density diverges.

Forwards: no collapse into a singularity – the winding/momentum minimum (appendix of Doc. 312) has $V'' > 0$.

What Doc. 312 left open is the *history*: observations show a clear thermal sequence. How does a gradient live on a circle? That is D(ii), and here it is computed, not narrated.

Chapter B

Preliminary: the fractal path correction

All distances in this document are *light paths*. But FFGFT space is not smooth-affine; it carries the fractal dimension $D_f = 3 - \xi$ (A040). A traversed path accumulates an elongation over it, so the smoothly computed distance is too large. Before any numbers, therefore, the question of where this factor applies.

B.1 The factor

A040 supplies it, and *derived*, not stipulated:

$$K_{\text{frak}} = \left(\frac{D_f^{\text{eff}}}{3}\right)^{D_f^{\text{eff}}/2} = 1 - 100\,\xi = 0.98665, \qquad D_f^{\text{eff}} \approx 2.973. \tag{B.1}$$

D_f^{eff} is the unique value for which two independent derivations of m_e/m_μ agree – the factor 100 is thus a derived quantity, *not* a decade count and not something adjustable to a desired result. Its uncertainty is quantified: $n = 100 \pm 2$ (register 190, R67), i.e. $\pm 0.027\,\%$ in the effect. The path elongation is $1/K_{\text{frak}} - 1 = 1.35\,\%$.

B.2 The assignment rule

A040 restricts sharply [B]: K_{frak} concerns *absolute* values, never ratios *of the same level* – there it cancels. What "same level" means is decided by the distinction of Doc. 311:

Quantity	Level	K_{frak}
Light path η_0, $\chi(z)$	unrolled (traversed)	applies
L^*, R_H, τ_c	rolled up (topological)	—
path / structure ($\eta_0/\pi R_H$)	mixed	**applies**
path / path (χ_{rec}/η_0)	same level	cancels
angles, dimensionless ratios	same level	cancels

The reason is conceptual, not numerical: a light path exists only in the *unrolled* representation – in the rolled-up view there is no "path to the antipode", only the cycle position θ. R_H, by contrast, follows from $H_0 = (\pi/2)c\xi^{10}/\bar{\lambda}_e$, a topological structural relation. So $\eta_0/(\pi R_H)$ compares something traversed with something topological and is *not* a same-level ratio.

Control test: $\chi_{\text{rec}}/\eta_0 = 0.98024$ is identical with and without the correction – two light paths, the factor cancels as required. The rule is thus not freely applicable; it has a testable invariant.

Declaration: this assignment is a conceptual stipulation, and it improves the numbers. Both belong stated side by side. In its favour: the direction of the effect is physically compelled (a winding path covers less smooth distance in the same travel time – no choice), and the level distinction of Doc. 311 was formulated independently and earlier. Nothing known speaks against it, but it is not proven. All following numbers are therefore carried in both versions.

Chapter C

The antipode finding [K|P20]

C.1 Where does the history sit on the cycle?

Position on the cycle: $\theta = \chi/R_H$, with χ the light path distance (the $z \to \chi$ assignment takes the measured ΛCDM map; only dimensionless ratios enter, Doc. 267). The antipode lies at $\theta = \pi$, i.e. $\chi = L^*/2 = \pi R_H = 14.09$ Gpc. Computed:

	fraction of $L^*/2$		
	smooth	fractal	
Recombination ($z = 1090$)	0.992	0.979	$-2.1\,\%$
Particle horizon ($z \to \infty$)	1.012	**0.9986**	$-$**0.14** $\%$

The **particle horizon** – the formal boundary of causal contact and hence the actual "beginning" – lands within $0.14\,\%$ of the antipode with the fractal correction, against $1.2\,\%$ in the smooth calculation. That is an improvement by a factor of 9 with nothing adjusted. The remainder is, however, five times the K_{frak} uncertainty ($\pm 0.027\,\%$), so not an exact hit.

The **scattering wall** thereby moves from 0.992 to 0.979: the secondary finding "recombination at the antipode" becomes *weaker*. Both statements cannot be exact simultaneously – they differ by the invariant factor $\chi_{\text{rec}}/\eta_0 = 0.980$.

z	θ/π smooth	θ/π fractal	$m(z)/m_0$
0.5	0.140	0.138	0.667
1	0.243	0.240	0.500
3	0.465	0.459	0.250
10	0.689	0.680	0.091
100	0.917	0.905	0.0099
1090	0.992	0.979	0.00092

Finding: the entire observable history – up to $z \to \infty$ – fills one *half-cycle* to $1.2\,\%$. The last-scattering surface sits at the antipode ($99.2\,\%$), the formal "beginning" $1.2\,\%$ behind it. The "big bang" is thereby a *place on the cycle* – the antipodal region, in the mass-drift reading the region of smallest masses and slowest clocks – and not an event before time.

C.2 Honest placement of the coincidence [S]

In ΛCDM language the same fact is the known quasi-coincidence $\eta_0 H_0/c \approx \pi$ (here 3.18 against 3.14, i.e. $1.2\,\%$) – there an accident of the measured Ω values. In the compact reading it becomes a structural statement: *the scattering wall sits at the antipode.* For it to be more than a re-labelled coincidence, the location would have to follow forward. The fractal path correction is a step in exactly that direction: independently derived (A040), not fitted to this result, and it reduces the particle horizon's distance from the antipode from $1.2\,\%$ to $0.14\,\%$. The location is not thereby fully derived – the remainder is five times the R67 uncertainty, and why the matter content fills exactly the half-cycle remains open. Status [S], noted as a derivation target.

Chapter D

The smooth closure [K]

D.1 The profile and its endpoint slope

Periodicity requires the mass drift $m(\theta)$ to close continuously on the circle. The simplest closure is an *even* profile about the antipode: minimum at $\theta = \pi$, symmetric return on the other half. It is smooth if $dm/d\theta \to 0$ at the turning point.

In the radiation era $H \simeq H_0\sqrt{\Omega_r}(1+z)^2$, hence analytically

$$\eta_0 - \chi = \frac{c}{H_0\sqrt{\Omega_r}}\frac{1}{1+z} \qquad \Longrightarrow \qquad \left|\frac{dm}{d\theta}\right|_{\text{end}} = \sqrt{\Omega_r} \approx 0.0096.$$

Finding: the observed profile runs into the endpoint with slope ~ 0.01 – the smoothness condition of an even periodic closure is met to $1\,\%$. The deformation required relative to the measured history is a $1\,\%$ effect at the outermost edge.

D.2 Why this is not self-deception

The return half – where the mass drift would have to rise again – is *exactly* the half behind the scattering wall: photonically unobservable in principle. That is suspiciously convenient at first sight, but it is not a choice of this document; it is a consequence of the antipode finding: the wall *lies* at $0.992\,\pi$. The closure therefore contradicts no single photon observation – and it is nevertheless not untestable (Ch. E).

The climate-zone picture from the Doc. 312 discussion thereby becomes quantitative: the gradient is the map *along* the cycle (like climate zones along a meridian circle), not an evolution *out of* a beginning – and the “hottest” point of the map is the antipode.

Chapter E

The entropy obstruction: the core of D(ii) [K]

E.1 Periodicity is not generic

The entropy budget of the observable volume is $S \sim 10^{104}\, k_B$ (dominated by supermassive black holes; photons: $10^{89}\, k_B$). Poincaré recurrence of a generic state takes $\sim e^{S/k_B} \sim 10^{10^{104}}$ dynamical times – against $\tau_c \sim 10^{18.5}$ s. The discrepancy is $\sim 10^{104}$ *orders of magnitude in the exponent*: exact τ_c periodicity cannot be expected dynamically.

Consequence: periodicity is a **boundary condition** – a selection of periodic solutions from the solution space, not an outcome of evolution. It belongs to the same category as the choice of the T^4 topology itself ([STIPULATION] in the sense of Doc. 311) and must be booked as such.

E.2 Fine- and coarse-grained, separated

Fine-grained [K]: under unitary evolution the fine-grained entropy is constant; nothing microscopic forbids an exactly periodic solution. The boundary condition is admissible.

Coarse-grained [open]: the second law is a statement about coarse-grained entropy relative to an observer. On the return half it must decrease – which is observer-relative possible (reversal of the correlation direction; no one *there* would see a violation, since the thermodynamic arrow turns with it), but unproven. **Exactly this is the sharpened core of D(ii):** not the field periodicity (which is consistent as a boundary condition, Ch. C), but the proof that a coarse-grained entropy closure is compatible with the observed structure formation. Until then: open, without embellishment.

Chapter F

Hawking radiation on the time cycle

Does Hawking radiation fit this picture? Yes – better than before, because the compact time direction (Doc. 312) has quietly supplied its foundation. At the same time it sharpens the entropy core of D(ii). Both computed (`ffgft_313_hawking_zyklus.py`).

F.1 One principle, three faces [K]

The three famous "temperatures out of nothing" – Unruh, Gibbons–Hawking, Hawking – are three separate miracles with three derivations in the textbooks. Behind them stands *one* KMS rule: where the time structure is periodic, an observer reads temperature, $T = \hbar/(k_B\tau)$:

System	Period τ	$T = \hbar/(k_B\tau)$
Cosmos (compact time)	$2\pi/H_0 = 2.9 \times 10^{18}$ s	2.63×10^{-30} K
Black hole ($M_\odot$)	$8\pi GM/c^3 = 1.24 \times 10^{-4}$ s	6.17×10^{-8} K
Unruh ($a = g$)	$2\pi c/a = 1.9 \times 10^8$ s	3.98×10^{-20} K

Doc. 312 *grounded* this rule for the cosmos: the Gibbons–Hawking temperature follows from the compactness of the time direction itself, without a horizon. Hawking radiation is therefore no foreign body and no extra postulate – it is the *local special case of the same rule*.

F.2 The membrane gets its thermometer [S]

The membrane picture (narrative Ch. 04: the hole *is* its fractal membrane, the radiation comes from the oscillating surface) so far had no answer to "why exactly this temperature?". Now it has one, in the language of the mass map: mass compresses the time structure ($\tilde{T} \cdot m = 1$ – the closer to the membrane, the slower the clocks), and at the membrane the local time structure becomes periodic with $\tau = 8\pi GM/c^3$. Short period, high temperature:

Object	T_H
PBH 10^{12} kg	1.2×10^{11} K
Earth	2.1×10^{-2} K
Sun	6.2×10^{-8} K
Sgr A*	1.5×10^{-14} K
SMBH $10^9 M_\odot$	6.2×10^{-17} K

The soap bubble of Ch. 04 has a thermometer, and the thermometer measures the clock map.

F.3 The family ladder [S]

Where does the ladder meet the cosmos? $T_H = T_{\text{GH}}$ requires $M = c^3/(4GH_0) = 4.66 \times 10^{52}$ kg – and its horizon lies at

$$r_s = \frac{2GM}{c^2} = \frac{R_H}{2} \quad \text{(exactly } 0.500\text{)},$$

at the same time exactly *half* the critical Hubble mass $c^3/(2GH_0)$. The temperature ladder runs from micro holes to the whole without a break: the cosmos is the largest family member.

F.4 What Hawking cannot do [K]

The honest friction with the cycle: evaporation times against $\tau_c = 91.9$ Gyr:

Object	t_{evap}	t_{evap}/τ_c
PBH 10^{12} kg	$10^{12.4}$ yr	$10^{1.5}$
Sun	$10^{67.3}$ yr	$10^{56.4}$
SMBH $10^9 M_\odot$	$10^{94.3}$ yr	$10^{83.4}$

Only primordial holes below the threshold mass

$$M^* = \left(\frac{\tau_c \hbar c^4}{5120\,\pi G^2}\right)^{1/3} = 3.3 \times 10^{11} \text{ kg} \quad (\sim \text{asteroid})$$

close via Hawking within one cycle. A solar-mass hole is 56 orders of magnitude too slow, an SMBH 83. The FFGFT correction from Ch. 04, $P = P_{\text{std}}(1 - \xi \ln(M/M_P))$, amounts to 0.6–$1.4\,\%$ and changes nothing in this finding.

Consequence for D(ii): stellar and supermassive holes – the carriers of the 10^{104} entropy budget of Ch. D – cannot disappear via Hawking on the cycle. The return half must dismantle them by another route. Hawking radiation thus fits the picture thermodynamically perfectly and at the same time *sharpens* the coarse-grained core of D(ii): it shows how hopelessly slow the only known return route would be.

F.5 Information: two languages, one statement [S]

Ch. 04 always said: the radiation correlates with the fractal membrane structure – nothing is lost ($S_{\text{BH}}(M_\odot) = 1.05 \times 10^{77}\,k_B$ of correlations). Ch. D of this document says: fine-grained entropy is constant under unitary evolution – periodicity is microscopically admissible. That is the same statement in two languages: information preservation locally (membrane) and globally (cycle) are consistent. What remains open is solely the *coarse-grained* closure – which was the core all along.

Chapter G

Test surfaces [S]

Photons end at the wall ($0.992\,\pi$). Two messengers originate at or behind the antipode:

Messenger	Decoupling	Position
CνB (neutrino background)	$z \sim 6 \times 10^9$	$\theta \approx 1.01\,\pi$
primordial GW	nominally $z \sim 10^{25}$	$\theta \approx 1.01\,\pi$

What the path correction costs here: fractally, the scattering wall lies at $0.979\,\pi$, clearly *below* the half-cycle – it does not intersect itself. On the one hand this makes the compact reading more compatible with the null results of the topology searches (the bound $(L^*/2)/D_{\mathrm{LSS}}$ rises from 1.008 to 1.022); on the other it removes a test signature: without self-intersection there are no matched circles and hence no $4.2'$ offset (Doc. 312). The price of the better numbers is a lost test opportunity – to be booked honestly, not concealed.

A *non-monotonic* signature there – the reversal of the mass drift beyond the antipode – thus remains the *only* surviving test surface of the closure. Practically beyond present measurement, but nameable; together with the $4.2'$ dipole offset (Doc. 312, torus rest frame) these are two concrete signatures of the compact reading.

Chapter H

The antipode condition as a prediction: Ω_m^* and the κ candidate

H.1 The inversion

Ch. B read the location of the scattering wall as a finding and honestly booked the coincidence $\eta_0 H_0/c \approx \pi$ as [S]. But the condition can be *inverted*: if the compact reading requires the particle horizon to close exactly at the antipode,

$$\eta_0 = \pi R_H \qquad \Longleftrightarrow \qquad \int_0^\infty \frac{dz}{E(z)} = \pi, \tag{H.1}$$

then Ω_m is no longer a free parameter. The condition solves uniquely (flat, $\Omega_r = 9.3 \times 10^{-5}$) to:

Without the fractal correction this gives $\Omega_m^* = 0.325$ ($+1.4\,\sigma$ vs Planck). With it, the condition reads $\int dz/E = \pi/K_{\text{frak}} = 3.1841$ and yields

$$\boxed{\Omega_m^* = 0.3139} \qquad \text{against Planck } 0.315 \pm 0.007 \quad \Rightarrow \quad -0.16\,\sigma. \tag{H.2}$$

The R67 uncertainty of the factor ($n = 100 \pm 2$) shifts this only between 0.3137 and 0.3141 – the prediction is robust against the one quantified freedom.

The cosmic sector thereby obtains, for the first time, a *prediction for a ΛCDM parameter* rather than a reinterpretation – falsifiable: sharper Ω_m measurements (DESI, Euclid) decide whether 0.325 holds.

What the comparison quantity is: Planck $\Omega_m = 0.315 \pm 0.007$ is not a raw datum but itself the output of a six-parameter ΛCDM fit to photon count rates – including ΛCDM transfer functions and fiducial cosmology in the analysis chains. Read precisely, the $1.4\,\sigma$ comparison is therefore: *condition against pipeline, not condition against nature.* This is the third inheritance level beside the SI anchor (Doc. 309) and the $E(z)$ form (above), and like those it is declared rather than hidden. It cuts both ways: decades of ΛCDM refinement produce robustness *and* pipeline lock-in – and where the compact reading inherits the form, it simultaneously inherits its fit successes for the shared parts. This is why the independent test points of this document deliberately sit in the minimally model-laden observable class: directions and shapes (the $4.2'$ dipole offset, non-monotonicity at the antipode) rather than fit parameters.

Declared inheritance: the calculation uses the ΛCDM *form* of $E(z)$ – including the $(1+z)^3$ term with full Ω_m. The prediction is therefore stated precisely: *the map must be shaped so that $\int dz/E = \pi$; in ΛCDM bookkeeping this reads* $\Omega_m = 0.325$. What carries the matter share of that bookkeeping is a separate question (Sec. F.3). Status hence [K|P20 $\times\, E(z)$ form] – the same inheritance structure as in Doc. 309, one level deeper.

H.2 Consolidation: obligation B and the coincidence become one condition

From Ω_m^* follows $\Omega_\Lambda^* = 0.675$ and hence a structural candidate for the so far open order-one factor of the closure scale (Doc. 312, obligation B):

$$\kappa^* = 3\,\Omega_\Lambda^* = 2.058 \qquad \text{against measured } \kappa = 2.12 \;\; (-2.9\,\%) \tag{H.3}$$

(smooth: 2.026, i.e. $-4.4\,\%$).

The same consolidation logic as with the 10^{123} problem (→ P20): two open items – obligation B and the [S] coincidence of Ch. B – are reduced to *one* condition. The cross-check shows this is not circular: $\kappa = 2.12$ would require $\Omega_m = 0.294$, and then $\eta_0 H_0/c = 3.268 \neq \pi$. Antipode exactness and today's Planck κ stand in a genuine $4\,\%$ tension – test contact, not a fit.

H.3 Placements and one new open item

Ω_Λ **loses its substance status:** the "dark-energy density" is the FLRW translation of the closure scale; if the antipode condition holds, it is predicted, not fitted. **Hubble tension:** the compact reading takes an unambiguous side with the CMB-anchored value ($H_0 = 66.8$ from the ξ^{10} chain against Planck 67.4); the local SH0ES excess would then have to be a map effect of the calibration [S].

Dark matter: two regimes, one open. Dark *matter* has deliberately not been mentioned so far – and the gap belongs named. *Galactically* (rotation curves, RAR) FFGFT needs no particle DM: there the transition scale a_0 carries the load (Doc. 308); this document says nothing about that, correctly so. *At the background level*, however, the $z \to \chi$ map silently inherits the CDM share: in ΛCDM bookkeeping, $\Omega_m^* = 0.325$ consists of baryons (≈ 0.049) plus cold dark matter (≈ 0.276). What carries this share in the compact reading – actual matter, or a property of the mass-drift profile $m(\theta)$ that books as matter in FLRW translation – is unresolved and is booked as a **second new open item**. It is independent of the galactic question and must not be conflated with it.

Sharpened open item: T_0 is twice bound but not derived. What sets the monopole temperature $T_0 = 2.725$ K? The corpus contains two bindings at very different levels. *(a) Relational (Doc. 279, canonical):* T_0 defines the Casimir length $L_\xi = (15\xi/\pi^2)^{1/4}\,\hbar c/(k_B T_0) = 100.24\,\mu$m with the ξ ratio $\rho_{\text{Casimir}}/\rho_{\text{CMB}} = \pi^2/(240\xi) = 308$; T_0 enters there as a *measured input*, and Doc. 279 itself warns that $m_e c^2/(k_B T_0)$ is a separate, non-geometric factor. *(b) Forward claim (Doc. 061, old layer):* $k_B T_0 = (16/9)\,\xi$ eV hits to $+0.92\,\%$ – but is not bookable by today's standard: the eV is a hidden SI anchor (the factor $1/510\,999$ to the Kammerton form is not structural, violating the Doc. 310 discipline), the factor $16/9$ is post-hoc, and the stated precision there ("$0.0002\,\%$") is wrong (actually $0.92\,\%$) – a correction-register candidate. *Conclusion:* dimensionlessly, $k_B T_0/(m_e c^2) = 4.6 \times 10^{-10}$ is not an integer power of ξ ($\log_\xi = 2.41$); by the R61 standard no numerology is attempted. T_0 remains *not derived forward* and is booked as an open question of the cosmic sector – relationally embedded, but without a carrier.

Chapter I

Status overview

Statement	Content	Status
Antipode finding	horizon at antipode: fractal $0.14\,\%$ (smooth $1.2\,\%$); wall $0.979\,\pi$	[K|P20]
Path correction	$K_{\text{frak}} = 1 - 100\xi$ applies to path/structure, cancels for path/path (test passed)	[K]/[S]
Coincidence placement	$\eta_0 H_0/c \approx \pi$: structural statement only after forward derivation	[S]
Smooth closure	endpoint slope $\sqrt{\Omega_r}/K \approx 0.0098$; closure met to $1\,\%$	[K]
Return half	= photonically unobservable half; no conflict with photon data	[K]
Periodicity	boundary condition (Poincaré discrepancy 10^{104} dex in the exponent), not dynamics	[K]
Entropy	fine-grained admissible; coarse-grained closure open – sharpened core of D(ii)	open
Hawking	local case of the same KMS rule (GH from Doc. 312); ladder meets cosmos at $r_s = R_H/2$	[K]/[S]
Evaporation	$M^* = 3.3 \times 10^{11}$ kg; $M_\odot$: 56 dex too slow – sharpens D(ii)	[K]
Test surfaces	CνB/GW from the antipode; non-monotonicity as signature	[S]
Ω_m^*	fractal 0.3139 ($-0.16\,\sigma$), smooth 0.325 ($+1.4\,\sigma$); $E(z)$ form inherited	[K|P20$\times E$]
κ^*	fractal 2.058 ($-2.9\,\%$), smooth 2.026: candidate for obligation B	[S]
Matched circles	wall without self-intersection: signature from Doc. 312 lapses	loss
DM (background)	$(1+z)^3$ share of the map: carrier unresolved (galactic: a_0, Doc. 308, separate)	open
T_0	relationally bound (Doc. 279: L_ξ, $\pi^2/240\xi$); the 061 relation is eV-anchored, not bookable; forward open	open

Balance: after this document, "no big bang" means precisely: the "beginning" is the antipode of the time cycle – a place, not an event; the observed history is one half of a map closable smoothly to $1\,\%$ (with the fractal path correction the particle horizon meets the antipode to $0.14\,\%$); and the remaining burden lies not with the fields but with coarse-grained entropy. D(ii) is thereby not settled but reduced to its hard core.

Registration: entry into Doc. 190 will be done separately.